

# Logistic Regression



# Definition

→ logistic regression is a method used for classification problems.

→ In this, we find probability that a given input belongs to a particular category.

→ In linear regression, we find

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \dots + \theta_n x_m$$

but it gives values in range

$$[-\infty, \infty]$$

→ we use sigmoid function that converts a real-value in  $[-\infty, \infty]$  into real-value in



$[0, 1]$

what is Sigmoid Function?

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

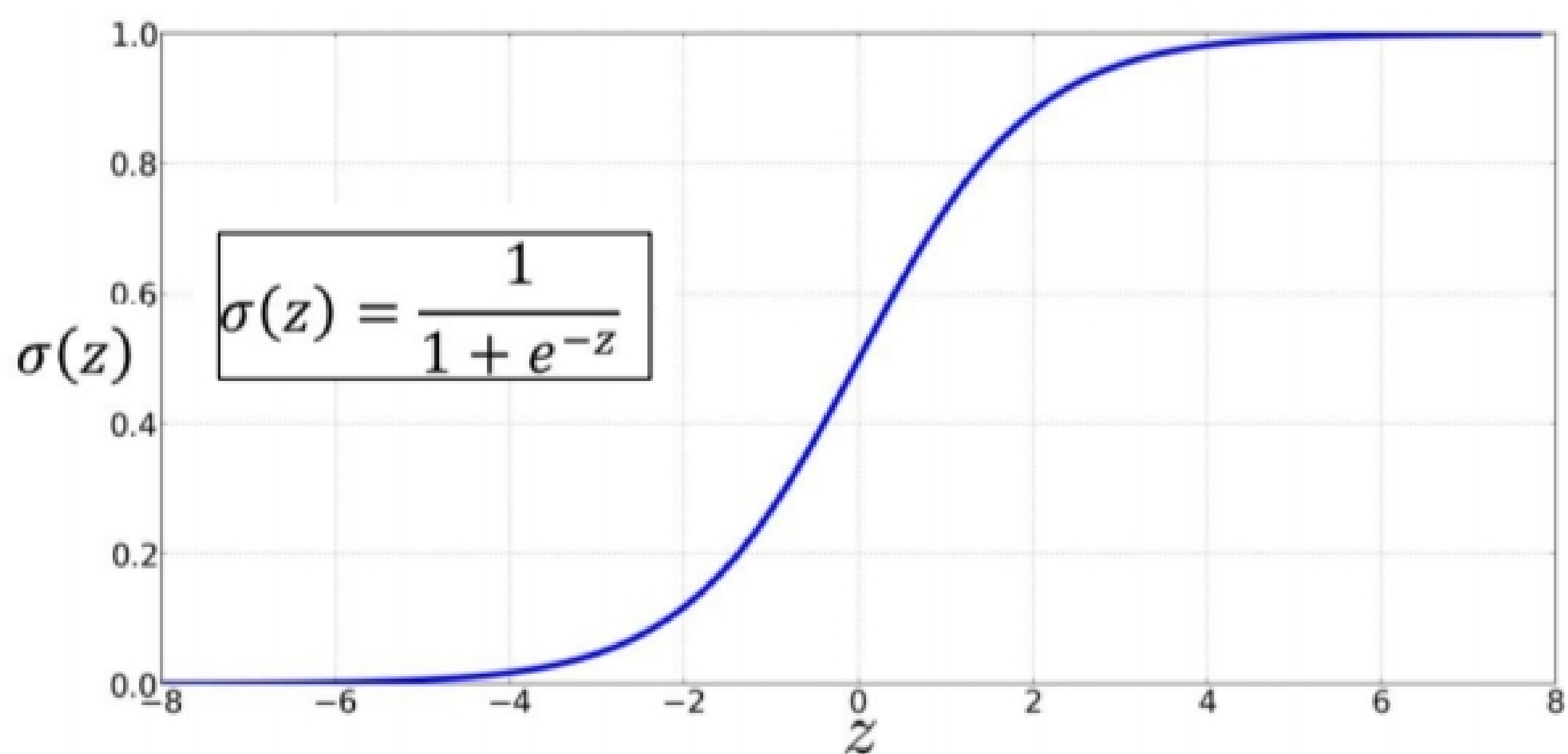
→ If  $x$  is +ve, it will make it -ve, hence denominator  $(1 + e^{-x})$  would become low, hence giving value more close to 1.

→ If  $x$  is -ve, denominator  $(1 + e^x)$  would become high, thus giving value more close to 0.



# why is it called "sigmoid"?

→ when we plot  $\sigma(x)$  for values of  $x$  in  $[-\infty, \infty]$ , it gives function curve of shape "S".



→ so we take sigmoid of linear function  $h_{\theta}(x)$  to get probabilities in range  $[0, 1]$

$$\sigma(h_{\theta}(x)) = \frac{1}{1 + e^{-h_{\theta}(x)}}$$



# Derivation of $\sigma(h_0(x))$

→ we model Odds to derive equation of logistic regression.

→ Odds are defined as:  
ratio of occurring of an event with its non-occurring

$$\text{odds} = \frac{p(x)}{1 - p(x)}$$

→ we assume that odds grow exponentially with linear score:

$$\text{linear score} = z = wX + b$$

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}, \quad b = b_0$$



$$\frac{p(x)}{1-p(x)} = e^z$$

If we take log on both sides

$$\log \left[ \frac{p(x)}{1-p(x)} \right] = \log e^z$$

✓  
called log-odds  
or logit

just for understanding

$$\frac{p(x)}{1-p(x)} = e^{wx+b}$$

$$p(x) = e^{wx+b} (1-p(x))$$

$$= e^{wx+b} - e^{wx+b} \times p(x)$$

$$p(x) + e^{wx+b} \times p(x) = e^{wx+b}$$



$$p(x) (1 + e^{\omega x + b}) = e^{\omega x + b}$$

$$p(x) = \frac{e^{\omega x + b}}{1 + e^{\omega x + b}}$$

Dividing denominator and numerator by  $e^{\omega x + b}$

$$\Rightarrow p(x) = \frac{e^{\omega x + b}}{e^{\omega x + b} + 1}$$

$$\left( \frac{1}{1 + e^{\omega x + b}} \right) / e^{\omega x + b}$$

$$= \frac{1}{\frac{1}{e^{\omega x + b}} + 1} = \frac{1}{1 + e^{-(\omega x + b)}}$$



$$\therefore p(x) = P(y=1/x)$$

- there is huge difference b/w

$p(x)$  and  $P(x)$ .

- $p(x)$  is short way of writing

$$P(y=1/x)$$

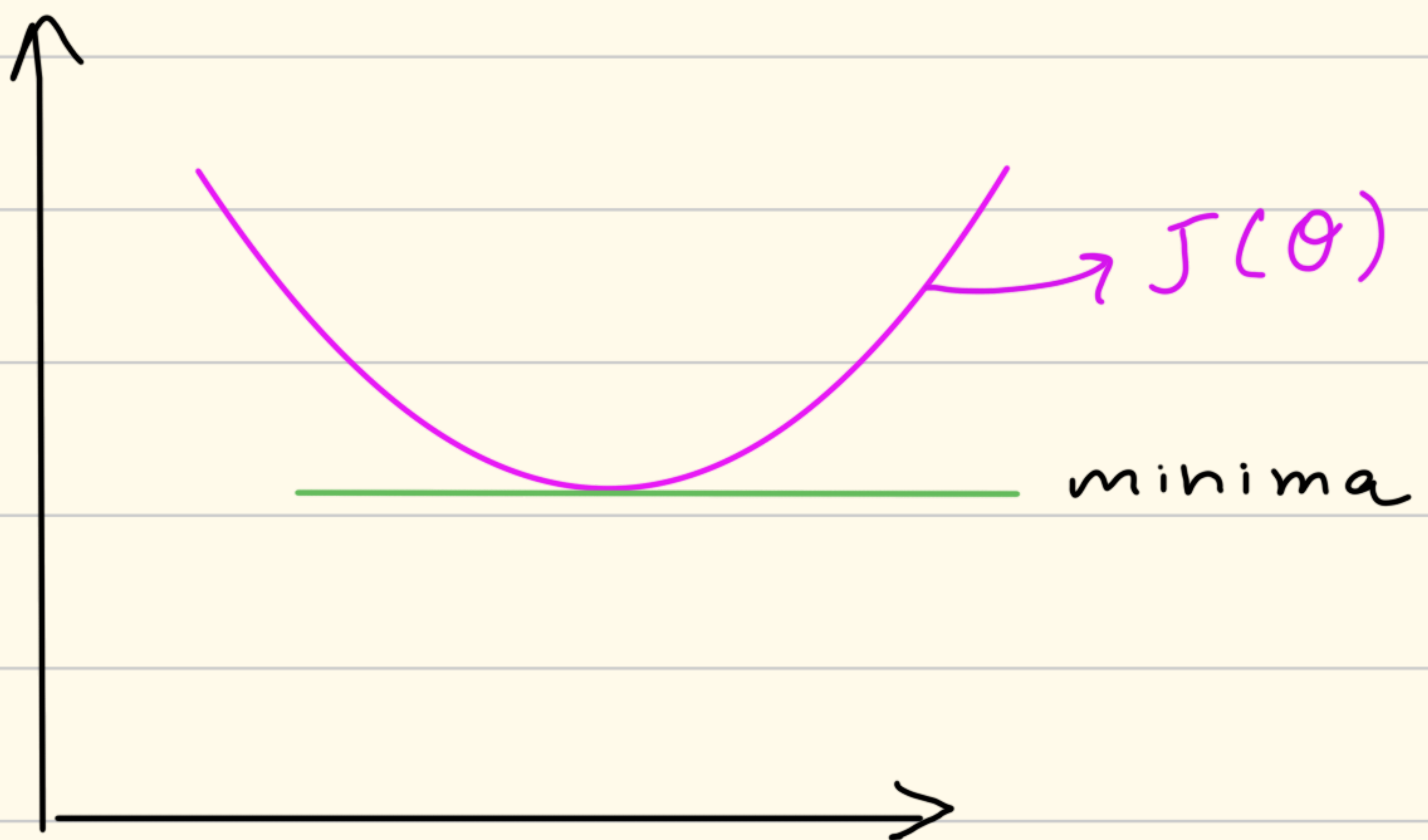


# How do we choose $\omega$ ?

→ In linear regression, we used cost function

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^i) - y^i)^2$$

- we minimized it and got  $\theta$  (or  $\omega$ )
- it was like this



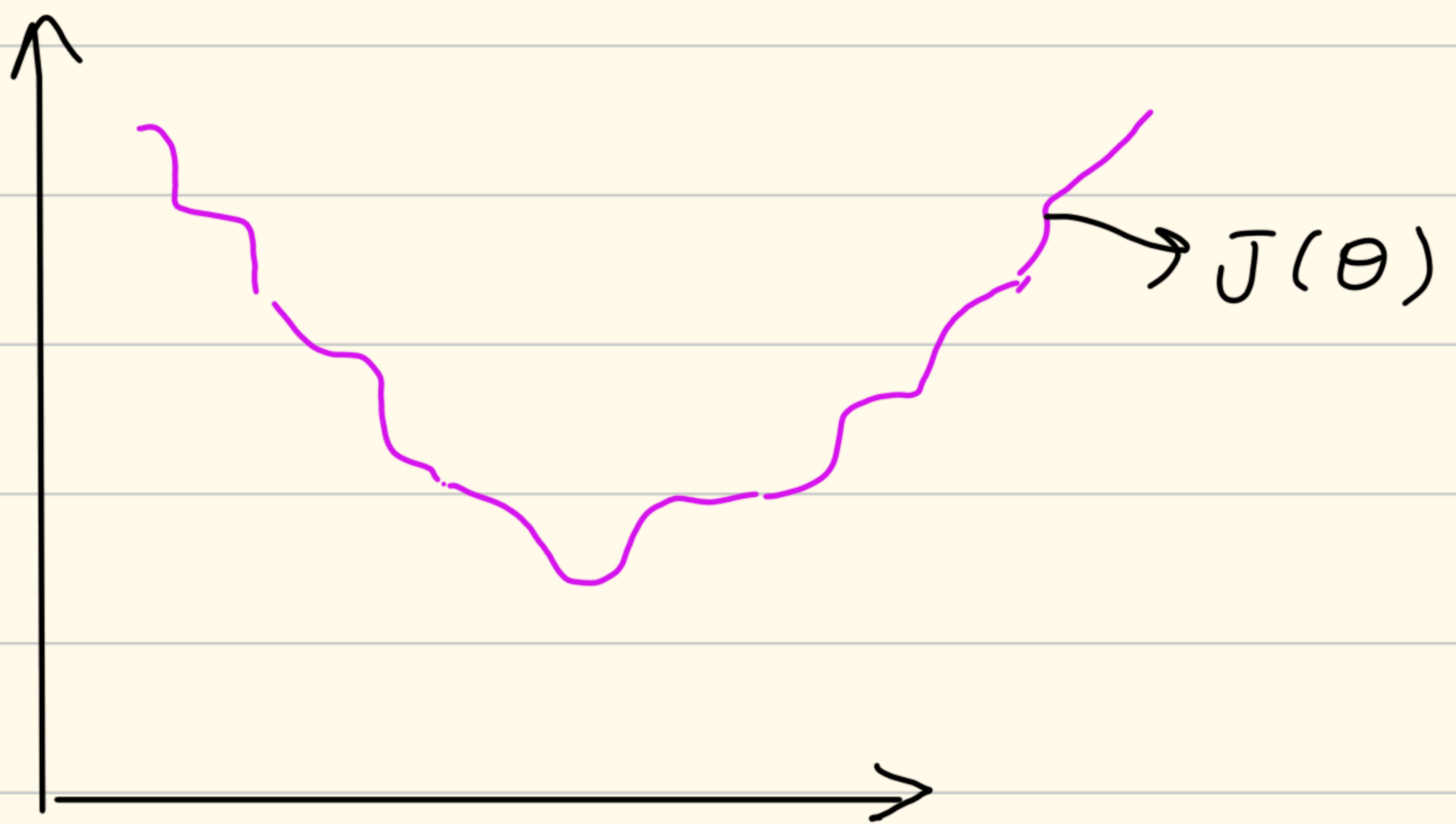
- function is convex, so we got one global minima.



→ If we map same thing in logistic regression like this:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^n \left( \frac{1}{1 + e^{-\theta^T x}} - y \right)^2$$

- This function is not convex:



so we can't use MSE here.

→ We use method of maximum likelihood here.



# Maximum Likelihood Method

→ Maximum likelihood method says that "choose parameters giving maximum likelihood for a dataset."

## Probability:

→ Given parameters, what is chance of observing data?

eg:

If coin is fair ( $p=0.5$  here is already known parameter), what is the probability of getting 7 heads in 10 tosses?



likelihood:

→ Given observed data, how plausible are different values.

eg:

If you observed 7 heads in 10 tosses what value of  $p$  (fairness of coin) best explains this?

→ So we find parameters  $w$  such the probability of dataset given these parameters is maximum.

$$L(w) = P(\text{Data} / w)$$

↙ likelihood function



So we need maximum of this function.

## Maximum Likelihood Method in Logistic Regression

→ we need to represent  $P(\text{Data} / \text{weights})$ .

Probability of each instance in Data can be represented using binomial representation.

means the binomial formula can give probability of 0 and 1 both classes.



$$P(y_i / x_i) = p_i^{y_i} (1 - p_i)^{1 - y_i}$$

$$\therefore p_i = \sigma(w^T x_i)$$

This is representation of probability of a single instance of dataset.

- if label is 1, then probability given by sigmoid would directly be probability of instance.
- If label is 0, then we do  $(1 - p)$  because sigmoid give for  $y=1$ , for  $y=0$  we do  $(1 - p)$



For complete dataset:

likelihood produced by weight  $w$ :

$$L(w) = \prod_{i=1}^m p_i^{y_i} (1 - p_i)^{1 - y_i}$$



likelihood function

we need to find  $w$  that maximizes likelihood of dataset.



→ It is hard to work with multiplication, so we take log on both sides to convert it into sum(s) form.

$$\log L(w) = \sum_{i=1}^m [y_i \log p_i + (1-y_i) \log (1-p_i)]$$

log-likelihood

→ we can't maximize  $\log L(w)$  easily and directly so we take its loss version and minimize that loss (effectively maximizing log likelihood)

$$\text{loss} = - \sum_{i=1}^m [y_i \log p_i + (1-y_i) \log (1-p_i)]$$



$\Rightarrow$

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[ y_i \log p_i + (1-y_i) \log (1-p_i) \right]$$

$$= -\frac{1}{m} \sum_{i=1}^m \left[ y_i \log \overbrace{\sigma(h(x))}^{p_i} + (1-y_i) \log \left( 1 - \overbrace{\sigma(h(x))}^{p_i} \right) \right]$$

we can write  $\sigma(h(x))$  as  $\hat{y}$

$$\Rightarrow J(\theta) =$$

$$-\frac{1}{m} \sum_{i=1}^m \left[ y_i \log \hat{y} + (1-y_i) \log (1-\hat{y}) \right]$$

$$\frac{d}{d\theta_0} J(\theta) =$$

$$-\frac{1}{m} \frac{d}{d\theta_0} \sum_{i=1}^m \left[ y_i \log \hat{y} + (1-y_i) \log (1-\hat{y}) \right]$$



Here chain rule would be implemented as

$$J \rightarrow \hat{y} \rightarrow h(x) \rightarrow \theta$$

$$\frac{d}{d\theta_0} J(\theta) = \underbrace{\frac{d}{d\hat{y}} J(\theta)} \times \underbrace{\frac{d}{dh(x)} \hat{y}} \times \underbrace{\frac{d}{d\theta_0} h(x)}$$

Let's find these one-by-one

(i)

$$\frac{d}{d\hat{y}} J(\theta) =$$

$$-\frac{1}{m} \frac{d}{d\hat{y}} \sum_{i=1}^m \left[ y \log \hat{y} + (1-y) \log (1-\hat{y}) \right]$$

$$= -\frac{1}{m} \sum_{i=1}^m \left[ y \times \frac{1}{\hat{y}} + (1-y) \times \frac{1}{(1-\hat{y})} (-1) \right]$$



$$= -\frac{1}{m} \sum_{i=1}^m \left[ \frac{y(1-\hat{y}) - \hat{y}(1-y)}{\hat{y}(1-\hat{y})} \right]$$

(ii)

$$\frac{d}{dh(x)} \hat{y} = \frac{d}{dh(x)} \left( \frac{1}{1+e^{-h(x)}} \right)$$

let's map it as  $\frac{d}{dx} \left( \frac{1}{1+e^{-x}} \right)$  and

Solve

$$\frac{d}{dx} \left( \frac{1}{1+e^{-x}} \right) = \frac{d}{dx} (1+e^{-x})^{-1}$$

using power rule:

$$= (-1)(1+e^{-x})^{-2} \frac{d}{dx} (1+e^{-x})$$

$$= (-1)(1+e^{-x})^{-2} \times (0+e^{-x}(-1))$$



$$= \frac{(-1)(e^{-x})(-1)}{(1+e^{-x})^2} = \frac{e^{-x}}{(1+e^{-x})^2}$$

$$= \frac{1}{(1+e^{-x})} \times \frac{e^{-x}}{(1+e^{-x})} = \sigma(x) \times \frac{e^{-x}}{(1+e^{-x})}$$

$$= \sigma(x) \times \frac{(1+e^{-x}-1)}{(1+e^{-x})}$$

$$= \sigma(x) \times \left( \frac{1+\cancel{e^{-x}}}{1+\cancel{e^{-x}}} - \frac{1}{1+e^{-x}} \right)$$

$$= \sigma(x) \times \left( 1 - \frac{1}{1+e^{-x}} \right)$$

$$= \sigma(x) [1 - \sigma(x)] \rightarrow \text{derivative of sigmoid}$$

$$\Rightarrow \frac{d}{dh(x)} \sigma(h(x)) =$$



$$\sigma(h(x)) [1 - \sigma(h(x))]$$

$$= \hat{y} [1 - \hat{y}]$$

(iii)

$$\frac{d}{d\theta_0} h(x) = \frac{d}{d\theta_0} (\theta_0 + \theta_1 x_1) = 1$$

From result of (i), (ii), (iii)

$$\frac{d}{d\theta_0} J(\theta) =$$

$$= -\frac{1}{m} \sum_{i=1}^m \left[ \frac{y(1-\hat{y}) - \hat{y}(1-y)}{\hat{y}(1-\hat{y})} \right] \times \hat{y}(1-\hat{y})$$

$$= -\frac{1}{m} \sum_{i=1}^m [y(1-\hat{y}) - \hat{y}(1-y)]$$



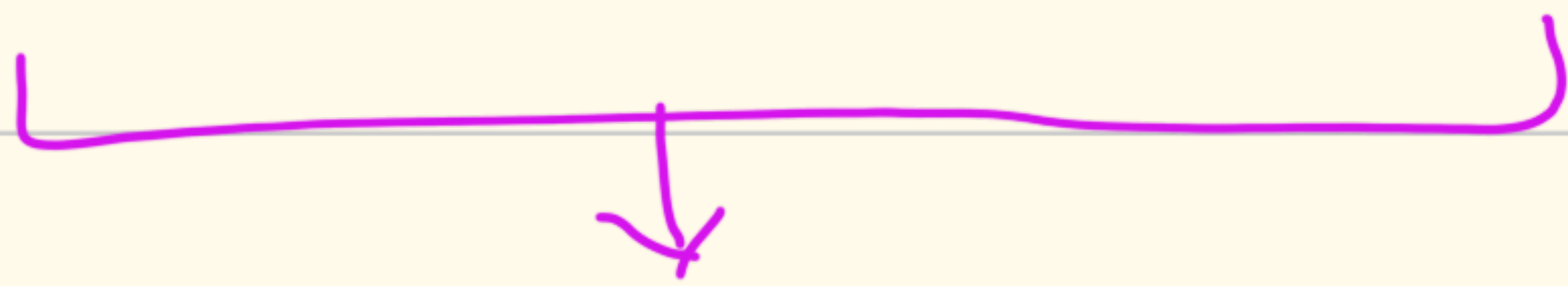
$$= -\frac{1}{m} \sum_{i=1}^m \left[ y - \cancel{y\hat{y}} - \hat{y} + \cancel{y\hat{y}} \right]$$

$$= -\frac{1}{m} \sum_{i=1}^m [y - \hat{y}]$$

$$\frac{dJ(\theta)}{d\theta_0} = \frac{1}{m} \sum_{i=1}^m [\hat{y}^i - y^i]$$

$$\frac{dJ(\theta)}{d\theta_1} =$$

$$\frac{dJ(\theta)}{d\hat{y}} \times \frac{d\hat{y}}{dh(x)} \times \frac{dh(x)}{d\theta_1}$$



These would be same as

extracted for  $\frac{dJ(\theta)}{d\theta_0}$  above



$$\frac{d}{d\theta_1} h(x) = \frac{d}{d\theta_1} (\theta_0 + \theta_1 x_1) = x_1$$

$\Rightarrow$

$$\frac{d}{d\theta_1} J(\theta) = \frac{1}{m} \sum_{i=1}^m [\hat{y}^i - y^i] x_1$$

General Rule:

$$\frac{d}{d\theta_j} J(\theta_1, \dots, \theta_n) =$$

$$\frac{1}{m} \sum_{i=1}^m [\hat{y}^i - y^i] x_j^i$$



Log Loss is also called Binary Cross Entropy?

Entropy:

$$\text{Entropy} = - \sum_x p(x) \log(p(x))$$

entropy = average uncertainties in system. (Here  $p(x)$  is basically weight of  $x$ .)

Cross-Entropy:

true prob distribution


$$\text{cross-entropy} = - \sum_x p(x) \log Q(x)$$

$\therefore Q(x)$  = model's prediction values

→ Cross entropy shows how



uncertain predictions are.

→ average uncertainties in model's predictions.

if we have two class and we use one-hot encoding for  $p(x)$  like for  $y=0$

$$p = [0, 1]$$

one-hot encoding

cross entropy =

$$- \left[ \overset{y}{\uparrow} (0) \log(p(0)) + \overset{(1-y)}{\uparrow} (1) \log(p(1)) \right]$$

↓

so here it becomes same as log loss.

⇒ Log loss is also called



binary cross entropy means  
log loss is special case of  
cross entropy.



# Binomial Logistic Regression

## Example

→ we will be given with input features ( $x$ s) and output feature  $y$  (two classes as it's binomial).

| $x$ (study hours) | $y$ (pass) |
|-------------------|------------|
| 0.5               | 0          |
| 1                 | 0          |
| 1.5               | 0          |
| 2                 | 0          |
| 2.5               | 0          |
| 3                 | 1          |
| 3.5               | 1          |
| 4                 | 1          |
| 4.5               | 1          |
| 5                 | 1          |



$$\theta_0 = -0.898$$

$$\theta_1 = 0.352$$

$$LR = 0.1$$

we need to find  $\theta_{0(new)}$  and  $\theta_{1(new)}$  to find

$$\sigma(h_{\theta}(x)) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

then we can use this  $\sigma(h_{\theta}(x))$  to predict classes.

we know that:

$$\theta_0 = \theta_0 - (LR) \times \frac{1}{m} \sum_{i=1}^m [\hat{y}_i - y_i]$$

$$\theta_j = \theta_j - (LR) \times \frac{1}{m} \sum_{i=1}^m [\hat{y}_i - y_i] x_j^i$$



| $h_{\theta}(x)$<br>$\theta_0 + \theta_1 x$ | $\hat{y}$<br>$\sigma(h_{\theta}(x))$ | $\hat{y} - y$ | $(\hat{y} - y)x$ |
|--------------------------------------------|--------------------------------------|---------------|------------------|
| -0.722                                     | 0.3270                               | 0.3270        | 0.1635           |
| -0.546                                     | 0.3668                               | 0.3668        | 0.3668           |
| -0.37                                      | 0.4085                               | 0.4085        | 0.61275          |
| -0.194                                     | 0.4516                               | 0.4516        | 0.9032           |
| -0.018                                     | 0.4955                               | 0.4955        | 1.2388           |
| 0.158                                      | 0.5394                               | -0.4606       | -1.3818          |
| 0.334                                      | 0.5827                               | -0.4173       | -1.4606          |
| 0.51                                       | 0.6248                               | -0.3752       | -1.5008          |
| 0.686                                      | 0.6651                               | -0.3349       | -1.5071          |
| 0.862                                      | 0.7031                               | -0.2969       | -1.4845          |

(0.1504) (-4.0498)

New values of  $\theta_0, \theta_1$

$$\theta_0 = (-0.898) - (0.1) \left( \frac{1}{10} \times (-4.0498) \right)$$



$$\theta_0 = -0.8965$$

$$\theta_1 = (0.352) - (0.1) \left( \frac{1}{10} \times (-4.0498) \right)$$

$$\theta_1 = 0.3925$$

$$\Rightarrow P(y=1|x) = \sigma(h_0(x))$$

$$= \frac{1}{1 + e^{-(-0.8965 + 0.3925x)}}$$

$$P(y=0|x) = 1 - P(y=1|x)$$